

Vector Calculus Exercises

Chapter 2.1

Exercise 11

Given the set of points $\{(x_i, y_i)\}_{i=1}^n$, and the function $f(m, b)$ defined as:

$$f(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2,$$

Minimize $f(m, b)$ using the method of least squares to find the best fit line $y = mx + b$ for the given data points.

We first compute the partial derivatives of $f(m, b)$ with respect to m and b , and set them to zero to find the critical points.:

$$\frac{\partial f}{\partial m} = -2 \sum_{i=1}^n x_i (y_i - mx_i - b) = 0,$$

$$\frac{\partial f}{\partial b} = -2 \sum_{i=1}^n (y_i - mx_i - b) = 0.$$

Solving the second equation for b gives:

$$b = \frac{1}{n} \sum_{i=1}^n y_i - m \frac{1}{n} \sum_{i=1}^n x_i = \bar{y} - m\bar{x},$$

where $\bar{x}$ and $\bar{y}$ are the means of the x_i and y_i values, respectively. Substituting this expression for b back into the first equation, we can solve for m :

$$-2 \sum_{i=1}^n x_i (y_i - mx_i - (\bar{y} - m\bar{x})) = 0.$$

Expanding and simplifying this equation leads to:

$$\sum_{i=1}^n x_i y_i - m \sum_{i=1}^n x_i^2 - \bar{y} \sum_{i=1}^n x_i + m\bar{x} \sum_{i=1}^n x_i = 0.$$

Then we can rearrange this to find m :

$$m = \frac{\sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i}.$$

Otherwise, we can express y_i in terms of the best fit line:

$$y_i = mx_i + b,$$

and form the matrix equation:

$$A \begin{bmatrix} m \\ b \end{bmatrix} = \mathbf{y},$$

where A is the matrix of x_i values and a column of ones, and $\mathbf{y}$ is the vector of y_i values.

This is an inconsistent system, although there exists a solution:

$$\begin{bmatrix} \hat{m} \\ \hat{b} \end{bmatrix},$$

such that:

$$\|Y - A\hat{X}\|^2 \leq \|Y - AX\|^2 \quad \text{for all } X,$$

where $\hat{Y} \in \text{Col}(A)$ and $Y - \hat{Y} \in (\text{Col}(A))^\perp$.

Then, $A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$ such that

$$a_i^T(Y - A\hat{X}) = 0 \quad \text{for } i = 1, 2.$$

The expression above can be rewritten as:

$$A^T(Y - A\hat{X}) = 0 \implies A^TY = A^TA\hat{X} \implies \hat{X} = (A^TA)^{-1}A^TY,$$

i.e., the normal equations for least squares. Writing

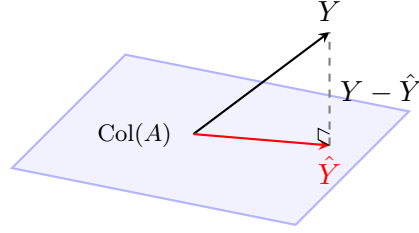
$$F(X) = f(m, b) = \|Y - AX\|^2 = (Y - AX)^T(Y - AX) = Y^TY - Y^TAX - X^TA^TY + X^TA^TAX,$$

we can compute the gradient of F as:

$$\nabla F(X) = -2A^TY + 2A^TAX.$$

Setting $\nabla F(X) = 0$ gives us the normal equations again:

$$A^TAX = A^TY \implies X = (A^TA)^{-1}A^TY.$$



Exercise 16

Given the function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ differentiable around -1 such that $h(-1) = 1$ and,

$$f(x, y) = h(xy) + 2h\left(\frac{y}{x}\right) - 4, \quad \text{for } x \neq 0,$$

find the gradient of f at the point $(-1, 1)$. Then, obtain the first-order Taylor expansion of f around the point $(-1, 1)$.

To find the gradient of f at the point $(-1, 1)$, we first compute the partial derivatives of f with respect to x and y . Using the chain rule, we have:

$$\frac{\partial f}{\partial x} = yh'(xy) - \frac{2y}{x^2} \cdot h'\left(\frac{y}{x}\right),$$

$$\frac{\partial f}{\partial y} = xh'(xy) + \left(\frac{2}{x}\right) \cdot h'\left(\frac{y}{x}\right).$$

Evaluating these derivatives at the point $(-1, 1)$, we get:

$$\frac{\partial f}{\partial x}(-1, 1) = 1 \cdot h'(-1) - \frac{2 \cdot 1}{(-1)^2} \cdot h'(-1) = h'(-1) - 2h'(-1) = -h'(-1),$$

$$\frac{\partial f}{\partial y}(-1, 1) = -1 \cdot h'(-1) + \left(\frac{2}{-1}\right) \cdot h'(-1) = -h'(-1) - 2h'(-1) = -3h'(-1).$$

Since $h'(-1) = 1$, the gradient of f at the point $(-1, 1)$ is:

$$\nabla f(-1, 1) = (-1 \quad -3).$$

The first-order Taylor expansion of f around the point $(-1, 1)$ is given by:

$$\mathcal{P}_{1,(-1,1)}(x, y) = f(-1, 1) + \nabla f(-1, 1) \cdot \begin{pmatrix} x + 1 \\ y - 1 \end{pmatrix}.$$

Substituting the values, we have:

$$\mathcal{P}_{1,(-1,1)}(x, y) = -1 - h'(-1)(x + 1) - 3h'(-1)(y - 1) = -1 - (x + 1) - 3(y - 1)$$

Chapter 2.2

Exercise 1

Given $f : \mathbb{R} \rightarrow \mathbb{R}$ differentiable, with $f'(x) \neq 0, \forall x \in \mathbb{R}$, prove it is one-to-one.

Proof. By contradiction. Assume f is not one-to-one. Then, there exist $t_0, t_1 \in \mathbb{R}$ such that $t_0 \neq t_1$ and $f(t_0) = f(t_1)$. By definition:

$$f'(t) = \lim_{t \rightarrow t_0} \frac{f(t) - f(t_0)}{t - t_0},$$

computing $f'(t_1)$ gives us:

$$f'(t_1) = \lim_{t \rightarrow t_0} \frac{f(t_1) - f(t_0)}{t_1 - t_0} = 0.$$

This would contradict the assumption that $f'(x) \neq 0$ for all $x \in \mathbb{R}$. Therefore, f must be one-to-one. $\square$

Given $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ a vector function defined as $F(x, y) = (e^x \cos y, e^x \sin y)$, prove that F is not one-to-one, even though its Jacobian determinant is non-zero for all $(x, y) \in \mathbb{R}^2$.

Proof. Taking two points $P_0 = \left(0, \frac{\pi}{4}\right)$ and $P_1 = \left(0, \frac{9\pi}{4}\right)$, we have:

$$F(P_0) = \left(e^0 \cos\left(\frac{\pi}{4}\right), e^0 \sin\left(\frac{\pi}{4}\right)\right) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right),$$

$$F(P_1) = \left(e^0 \cos\left(\frac{9\pi}{4}\right), e^0 \sin\left(\frac{9\pi}{4}\right)\right) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$

Thus, $F(P_0) = F(P_1)$ even though $P_0 \neq P_1$, proving that F is not one-to-one. However, the Jacobian determinant of F is given by:

$$\det J_F(x, y) = \det \begin{bmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix} = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x} > 0 \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Applying the Inverse Function Theorem, we can conclude that F is locally invertible at every point in $\mathbb{R}^2$, but it is not globally one-to-one. $\square$

If H is the inverse of F defined above, compute DH for $F(0, 0) = (1, 0)$.

Since $\det J_F(x, y) = e^{2x} > 0$ for all $(x, y) \in \mathbb{R}^2$, $M = \{(x, y) \in \mathbb{R}^2 : -2\pi < y < 2\pi\}$,

$$J_F(0, 0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \implies \det J_F(0, 0) = 1 \neq 0.$$

$$J_H(F_1, F_2) = \underbrace{J_F^{-1}(F_1, F_2)}_{\text{Inverse Function Theorem}} = (J_F(x, y))^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Exercise 2

Given the system of equations:

$$\begin{cases} u = x + xyz \\ v = y + xy \\ w = z + 2x + 3x^2 \end{cases}$$

solve for x, y, z in terms of u, v, w , near the point $(x, y, z) = (0, 0, 0)$.

Define the function $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ as:

$$F(x, y, z) = \begin{bmatrix} x + xyz \\ y + xy \\ z + 2x + 3x^2 \end{bmatrix}.$$

We need to find the inverse of that function (if possible). Applying the Inverse Function Theorem, we compute the Jacobian matrix of F :

$$J_F(x, y, z) = \begin{bmatrix} 1 + yz & xz & xy \\ y & 1 + x & 0 \\ 2 + 6x & 0 & 1 \end{bmatrix}.$$

Evaluating the Jacobian at the point $(0, 0, 0)$ gives us:

$$J_F(0, 0, 0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}.$$

The determinant of $J_F(0, 0, 0)$ is:

$$\det J_F(0, 0, 0) = 1 \cdot (1 \cdot 1 - 0 \cdot 0) - 0 \cdot (0 \cdot 1 - 0 \cdot 2) + 0 \cdot (0 \cdot 0 - 1 \cdot 2) = 1 \neq 0.$$

Since the determinant is non-zero, the Inverse Function Theorem guarantees that F is locally invertible around the point $(0, 0, 0)$. [...]

Exercise 3

Check that $z^3 + x(z - y) = 1$ defines $z = f(x, y)$ as an implicit function of class $\mathcal{C}^\infty$ near the point $(x, y, z) = (0, 1, 1)$.

We apply the Implicit Function Theorem to find z . Define the function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ as:

$$F(x, y, z) = z^3 + x(z - y) - 1.$$

We need to check if $F(0, 1, 1) = 0$ and if $J_F(0, 1, 1)$ has a non-zero entry in the column corresponding to z . Evaluating F at the point gives us:

$$F(0, 1, 1) = 1^3 + 0 \cdot (1 - 1) - 1 = 0.$$

Next, we compute the Jacobian of F :

$$J_F(x, y, z) = \begin{bmatrix} z - y & -x & 3z^2 + x \end{bmatrix} \implies J_F(0, 1, 1) = \begin{bmatrix} 0 & 0 & 3 \end{bmatrix}.$$

So we have x, y free variables, and z is the pivot variable. Hence, we can apply the Implicit Function Theorem to conclude that there exists a function $z = f(x, y)$ of class $\mathcal{C}^\infty$ defined in a neighborhood of $(0, 1, 1)$ and satisfying $F(x, y, f(x, y)) = 0$ for all (x, y) in that neighborhood.

Study if $(0, 1)$ is a local maximum, minimum, or saddle point of f .

Using implicit differentiation, we can compute the first and second partial derivatives of f at the point $(0, 1)$. The first partial derivatives are given by:

$$\frac{\partial F}{\partial x} = 3[z(x, y)]^2 \cdot \frac{\partial z}{\partial x} + (z(x, y) - y) + x \cdot \frac{\partial z}{\partial x} = 0,$$

$$\frac{\partial F}{\partial y} = 3[z(x, y)]^2 \cdot \frac{\partial z}{\partial y} - x + x \cdot \frac{\partial z}{\partial y} = 0.$$

Evaluating these at the point $(0, 1)$ gives us:

$$\frac{\partial F}{\partial x}(0, 1) = 3 \cdot 1^2 \cdot \frac{\partial z}{\partial x}(0, 1) + (1 - 1) + 0 \cdot \frac{\partial z}{\partial x}(0, 1) = 3 \cdot \frac{\partial z}{\partial x}(0, 1) = 0 \implies \frac{\partial z}{\partial x}(0, 1) = 0,$$

$$\frac{\partial F}{\partial y}(0, 1) = 3 \cdot 1^2 \cdot \frac{\partial z}{\partial y}(0, 1) - 0 + 0 \cdot \frac{\partial z}{\partial y}(0, 1) = 3 \cdot \frac{\partial z}{\partial y}(0, 1) = 0 \implies \frac{\partial z}{\partial y}(0, 1) = 0.$$

Thus, $(0, 1)$ is a critical point of f . To determine its nature, we need to compute the second partial derivatives and evaluate the Hessian matrix at this point.

$$\frac{\partial^2 F}{\partial x^2} = 6z \cdot \left(\frac{\partial z}{\partial x} \right)^2 + 3z^2 \cdot \frac{\partial^2 z}{\partial x^2} + 2 \cdot \frac{\partial z}{\partial x} + x \cdot \frac{\partial^2 z}{\partial x^2} = 0,$$

$$\frac{\partial^2 F}{\partial y^2} = 6z \cdot \left(\frac{\partial z}{\partial y} \right)^2 + 3z^2 \cdot \frac{\partial^2 z}{\partial y^2} + x \cdot \frac{\partial^2 z}{\partial y^2} = 0,$$

$$\frac{\partial^2 F}{\partial x \partial y} = 6z \cdot \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} + 3z^2 \cdot \frac{\partial^2 z}{\partial x \partial y} - 1 + x \cdot \frac{\partial^2 z}{\partial x \partial y} = 0.$$

Evaluating these at the point $(0, 1)$ gives us:

$$\frac{\partial^2 F}{\partial x^2}(0, 1) = 0 + 3 \cdot 1^2 \cdot \frac{\partial^2 z}{\partial x^2}(0, 1) + 2 \cdot 0 + 0 \cdot \frac{\partial^2 z}{\partial x^2}(0, 1) = 3 \cdot \frac{\partial^2 z}{\partial x^2}(0, 1) \implies \frac{\partial^2 z}{\partial x^2}(0, 1) = 0,$$

$$\frac{\partial^2 F}{\partial y^2}(0, 1) = 0 + 3 \cdot 1^2 \cdot \frac{\partial^2 z}{\partial y^2}(0, 1) + 0 \cdot \frac{\partial^2 z}{\partial y^2}(0, 1) = 3 \cdot \frac{\partial^2 z}{\partial y^2}(0, 1) \implies \frac{\partial^2 z}{\partial y^2}(0, 1) = 0,$$

$$\frac{\partial^2 F}{\partial x \partial y}(0, 1) = 0 + 3 \cdot \frac{\partial^2 z}{\partial x \partial y}(0, 1) - 1 \cdot \frac{\partial^2 z}{\partial x \partial y}(0, 1) = 3 \cdot \frac{\partial^2 z}{\partial x \partial y}(0, 1) - 1 \implies \frac{\partial^2 z}{\partial x \partial y}(0, 1) = \frac{1}{3}.$$

The determinant of the Hessian matrix of f at the point $(0, 1)$ is given by:

$$\det H_f(0, 1) = \det \begin{bmatrix} \frac{\partial^2 z}{\partial x^2}(0, 1) & \frac{\partial^2 z}{\partial x \partial y}(0, 1) \\ \frac{\partial^2 z}{\partial x \partial y}(0, 1) & \frac{\partial^2 z}{\partial y^2}(0, 1) \end{bmatrix} = \det \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{bmatrix} = -\frac{1}{9} < 0,$$

which indicates that $(0, 1)$ is a saddle point of f .

Exercise 4

Given the system of equations:

$$\begin{cases} xz^3 + y^2u^3 - 1 = 0 \\ 2xy^3 + u^2z = 0 \end{cases}$$

find $z = g_1(x, y)$ and $u = g_2(x, y)$ as implicit functions.

Apply the Implicit Function Theorem to find z and u . Define the function $F : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ as:

$$f(x, y, z, u) = \begin{bmatrix} xz^3 + y^2u^3 - 1 \\ 2xy^3 + u^2z \end{bmatrix} = (0, 0).$$

so for our notation, we have $N = 4$ and $M = 2$. We compute the Jacobian matrix of f at a point $(0, 1, 0, 1)$:

$$J_f(x, y, z, u) = \begin{bmatrix} z^3 & 2yu^3 & 3xz^2 & 3y^2u^2 \\ 2y^3 & 6xy^2 & u^2 & 2uz \end{bmatrix}$$
$$J_f(0, 1, 0, 1) = \begin{bmatrix} 0 & 2 & 0 & 3 \\ 2 & 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 3 \end{bmatrix}.$$

We have x, y as free variables, and z, u as pivot variables. Looking for $z = g_1(x, y)$ and $u = g_2(x, y)$, we define $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ as:

$$F(x, y, z, u) = \begin{bmatrix} x \\ y \\ xz^3 + y^2u^3 - 1 \\ 2xy^3 + u^2z \end{bmatrix}.$$

In terms of the Implicit Function Theorem, that is the splitting of the variables. The Jacobian matrix of F is given by:

$$J_F(x, y, z, u) = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline z^3 & 2yu^3 & 3xz^2 & 3y^2u^2 \\ 2y^3 & 6xy^2 & u^2 & 2uz \end{array} \right].$$

The determinant of the Jacobian matrix of F is:

$$\det J_F(x, y, z, u) = \det \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ z^3 & 2yu^3 & 3xz^2 & 3y^2u^2 \\ 2y^3 & 6xy^2 & u^2 & 2uz \end{bmatrix} = \det \begin{bmatrix} 3xz^2 & 3y^2u^2 \\ u^2 & 2uz \end{bmatrix} = 6xz^3u - 3y^2u^4.$$

Then, we know the existence of an inverse function so that:

$$\begin{aligned} x &= F_1 & z &= g_1(F_1, F_2, F_3, F_4) = g_1(x, y, F_3, F_4) \\ y &= F_2 & u &= g_2(F_1, F_2, F_3, F_4) = g_2(x, y, F_3, F_4) \end{aligned}$$

Chapter 2.3

Exercise 3

Given the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $f(x, y) = x^2 + y^2$ on $A = \{xy = 1\}$, find the minimum of f on A .

To find the minimum of f on the set A , we can use the method of Lagrange multipliers. We define the constraint function $g(x, y) = xy - 1$. The Lagrangian function is given by:

$$\mathcal{L} = f(x, y) - \lambda g(x, y) = x^2 + y^2 - \lambda(xy - 1).$$

We need to compute the gradients and set them equal to zero:

$$\nabla \mathcal{L} = \begin{bmatrix} 2x - \lambda y \\ 2y - \lambda x \\ -(xy - 1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This gives us the system of equations:

$$\begin{cases} 2x - \lambda y = 0 \\ 2y - \lambda x = 0 \\ xy - 1 = 0 \end{cases}$$

From the first two equations, we can express λ in terms of x and y :

$$\lambda = \frac{2x}{y} = \frac{2y}{x} \implies 2x^2 = 2y^2 \implies x^2 = y^2.$$

Substituting $y = \pm x$ into the constraint $xy = 1$ gives us:

$$\begin{cases} x \cdot x = 1 \implies x = 1, y = 1 \\ x \cdot (-x) = 1 \implies x = -1, y = -1 \end{cases}$$

Evaluating f at these points gives us:

$$f(1, 1) = 1^2 + 1^2 = 2,$$

$$f(-1, -1) = (-1)^2 + (-1)^2 = 2.$$

Thus, the minimum of f on the set A is 2, achieved at the points $(1, 1)$ and $(-1, -1)$.

